Research Experiment Results

1. System-Level Objective Evaluation

These results correspond to the main engineering objectives of the proposed hierarchical federated multimodal system. The wearable accuracy result is treated as inconclusive where its confidence interval crosses the target threshold.

Experiment / Objective	Target	Clinical	Wearable	Status
Uplink reduction	≥ 50%	90.82%	96.08%	Pass
Round latency improvement	> 0%	16.93%	21.68%	Pass
Composed privacy budget	≤ 1	0.970	0.966	Pass
Accuracy loss	≤ 2 pp	−0.92 pp	1.97 pp	Clinical pass: wearable inconclusive
P95 inference latency	< 100 ms	4.58 ms	1.22 ms	Pass
Successful updates	> 99%	100.00%	99.94%	Pass
Energy saving	≥ 20%	72.30%	76.84%	Pass

2. Multimodal Fusion Experiment

The multimodal result is treated as a system-setting result rather than the central scientific claim. Fusion is beneficial on VitalDB but provides little additional headroom on WEAR because the stronger unimodal model is already near ceiling.

Dataset	Best unimodal AUROC	Fusion AUROC	Fusion gain	Interpretation
VitalDB	0.8283	0.8653	+0.0370	Meaningful fusion gain
WEAR	0.9976	0.9980	+0.0004	Ceiling: fusion adds little

3. Privacy-Unit / k-Scaling Experiment

This is the main scientific experiment. k denotes the number of records contributed per patient. The observed record-level versus patient-level utility gap increases after k=1, while the patient arm remains approximately flat.

k (records/patient)	Record/sample arm AUPRC	Patient arm AUPRC	Gap	p-value
1	0.5015	0.5067	−0.0052	0.75
2	0.5590	0.5084	+0.0507	0.0017
4	0.5924	0.5092	+0.0832	0.0015

8	0.5955	0.5089	+0.0866	0.0049

Interpretation: k=1 provides a built-in null control because record and patient are the same privacy unit. For k>1, the record-level arm rises while the patient-level arm stays approximately flat. This supports an observed records-per-patient dependence: external validation is still required before calling it a universal law.

4. Membership-Inference Audit

The original MIA hypothesis predicted a substantially larger patient-level attack advantage for record level DP. The paired LiRA results did not establish that large effect. A positive control demonstrated that the attack can detect leakage.

Arm	MIA AUC	Original interpretation
Non-private control	0.6230	Positive control: attack signal present
Record-level DP, ε≈1	0.5477	Small signal: large predicted effect not established
Patient-level DP, ε≈1	0.5224	Near chance: large predicted effect not established

5. MIA Difficulty-Floor Control

A new control corrects an important evaluation issue: members and non-members differ in intrinsic difficulty. A shadow-only attacker without the target model establishes a difficulty floor. Attack performance should therefore be interpreted relative to this floor rather than only against AUC=0.5.

Arm	Attack AUC	Difficulty floor	Margin over floor
Positive control	0.6254	0.5626	+0.0627
Record DP, ε≈1	0.5477	0.5252	+0.0226
Patient DP, ε≈1	0.5224	0.5167	+0.0057
Record DP, ε≈16	0.5465	0.5743	−0.0278
Record DP, ε≈16	0.5630	0.5259	+0.0371

Interpretation: the positive control still clears its difficulty floor, supporting the validity of the attack. The ε≈1 private arms clear their floors only slightly, and the ε=16 record arm falls below its floor. This makes the refutation of the originally predicted large MIA effect stronger.

6. Privacy Budget / Training Configuration

Configuration	Key result	Interpretation
Patient-level matched-budget DP	ε ≈ 0.97 composed	Meets ε≤1 target
Training + conformal calibration	0.7705 + 0.2000 = 0.9705	Composition remains below 1
Patient-level upgrade	≈1.77× σ vs record-level	Stronger privacy unit has measurable optimization cost
Head-only DP	AUC 0.8742: much faster/stronger utility	Parameter-efficient private adaptation is promising

7. Ablation Experiments

Removed component	Observed result	Conclusion
Hierarchy	AUROC change −0.0058: CI spans zero	Hierarchy mainly supports systems efficiency
Privacy	AUROC change −0.0234: CI excludes zero	Privacy mechanism has measurable utility cost
Sparsification	~5× larger uplink	Compression materially reduces communication
Quantisation	~1.6× larger uplink	Quantisation contributes further payload reduction
Sparsification + quantisation	~8× payload reduction	Large communication benefit without measurable discrimination loss